

SQL Functions Assignment

Question 1 : Create a ranking of students based on score (highest first).

Answer:-

```
SELECT student_id, name, course, score, attendance, mentor, join_date, city
FROM Student_Performance
ORDER BY score DESC;
```

Question 2 : Show each student's score and the previous student's score (based on score order).

Answer:-

```
SELECT
    student_id,
    name,
    course,
    score,
    LAG(score) OVER (ORDER BY score DESC) AS previous_score
FROM Student_Performance
ORDER BY score DESC;
```

Question 3 : Convert all student names to uppercase and extract the month name from join_date.

Answer:-

```
SELECT
    UPPER(name) AS student_name_upper,
    course,
    score,
    attendance,
    mentor,
    join_date,
    city,
    MONTHNAME(join_date) AS join_month
FROM Student_Performance;
```

Question 4 : Show each student's name and the next student's attendance (ordered by attendance).

Answer:-

```
SELECT
    name,
    attendance,
    LEAD(attendance) OVER (ORDER BY attendance ASC) AS next_attendance
FROM Student_Performance
ORDER BY attendance ASC;
```

Question 5 : Assign students into 4 performance groups using NTILE().

Answer:-

```
SELECT
    student_id,
    name,
```

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```
course,  
score,  
NTILE(4) OVER (ORDER BY score DESC) AS performance_group  
FROM Student_Performance  
ORDER BY score DESC;
```

Question 6 : For each course, assign a row number based on attendance (highest first).

Answer:-

```
SELECT  
    student_id,  
    name,  
    course,  
    attendance,  
    ROW_NUMBER() OVER (PARTITION BY course ORDER BY attendance DESC) AS  
attendance_rank  
FROM Student_Performance  
ORDER BY course, attendance_rank;
```

Question 7 : Calculate the number of days each student has been enrolled (from join_date to today).
(Assume current date = '2025-01-01')

Answer:-

```
SELECT  
    student_id,  
    name,  
    course,  
    join_date,  
    DATEDIFF('2025-01-01', join_date) AS days_enrolled  
FROM Student_Performance  
ORDER BY days_enrolled DESC;
```

Question 8 : Format join_date as “Month Year” (e.g., “June 2023”).

Answer:-

```
SELECT  
    name,  
    course,  
    join_date,  
    DATE_FORMAT(join_date, '%M %Y') AS formatted_join_date  
FROM Student_Performance;
```

Question 9 : Replace the city ‘Mumbai’ with ‘MUM’ for display purposes.

Answer:-

```
SELECT  
    student_id,  
    name,
```

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```
course,  
score,  
attendance,  
mentor,  
join_date,  
REPLACE(city, 'Mumbai', 'MUM') AS display_city  
FROM Student_Performance;
```

Question 10 : For each course, find the highest score using FIRST_VALUE().

Answer:-

```
SELECT  
course,  
name,  
score,  
FIRST_VALUE(score) OVER (PARTITION BY course ORDER BY score DESC) AS  
highest_score_in_course  
FROM Student_Performance  
ORDER BY course, score DESC;
```
